A conservative receiver and the origin of a correlation scale

A finite harmonic receiver has a vanishing long-window action observable at fixed centre velocity. An increasing periodic chain instead has plateau $\Theta\sqrt{m/k}$ when its size tends to infinity first, with mode energy Θ and spring stiffness k. A common hard particle-speed ceiling on the stated phase preparation forces this plateau to close. The comparison locates the positive scale in the low-frequency weights and energy preparation.

1. Hamiltonian and invariant preparation

Let $x, p \in \mathbb{R}^n$, $n \geq 2$, $M = \operatorname{diag}(m_1, \ldots, m_n)$ with $m_i > 0$, and let K be a connected spring-network Laplacian. Thus $K = K^T \geq 0$ and $\ker K = \operatorname{span}\{\mathbf{1}\}$. The Hamiltonian and equations are

$$\mathcal{H} = \tfrac{1}{2}p^T M^{-1} p + \tfrac{1}{2}x^T K x, \qquad \dot{x} = M^{-1}p, \quad \dot{p} = -Kx.$$

Energy and total momentum are conserved. These linear equations give global smooth trajectories. Write $D = M^{-1/2}KM^{-1/2}$ and choose an orthonormal eigenbasis $e_0, e_1, \ldots, e_{n-1}$, with $e_0 = M^{1/2}\mathbf{1}/\sqrt{M_{\text{tot}}}$ and positive eigenvalues ω_j^2 for $j \geq 1$. Define canonical coordinates $q_j = e_j^T M^{1/2} x$, $P_j = e_j^T M^{-1/2} p$.

Fix centre velocity $V_0 = 0$ and centre position. Assign internal mode energies $E_j \geq 0$, and independent phases ϕ_j uniform on $[0, 2\pi)$. Then

$$q_j(t) = \frac{\sqrt{2E_j}}{\omega_j}\cos(\omega_j t + \phi_j), \qquad P_j(t) = -\sqrt{2E_j}\sin(\omega_j t + \phi_j).$$

This is an invariant probability ensemble on the internal phase tori, even when frequencies are commensurate. It is a preparation, not a mixing premise. The total energy is $E = \sum_j E_j$, so $|\dot{x}_i| \leq \sqrt{2E/m_i}$ for every trajectory. Choosing $E < \frac{1}{2}(\min_i m_i)c^2$ puts all particles below c in this frame. These are Newtonian springs with a speed-bounded preparation; propagation of signals and Lorentz covariance are separate model requirements.

2. Exact action observable and its limits (C047)

For tagged particle i, let $b_{ij} = e_{ji}/\sqrt{m_i}$, where e_{ji} denotes component i of vector e_j. Then $v_i(t) = \sum_{j\geq 1} b_{ij} P_j(t)$ has zero mean and covariance

$$C_i(t) = \sum_{j\geq 1} w_{ij}\cos(\omega_j t), \qquad w_{ij} = b_{ij}^2 E_j \geq 0.$$

Phase averaging gives this identity: distinct modes have zero cross covariance, and one phase has $\mathbb{E}[P_j(t)P_j(0)] = E_j\cos(\omega_j t)$. Integrating the covariance over the displacement square yields, for $\Delta > 0$,

$$\mathsf{h}_i(\Delta) = \frac{m_i}{\Delta}\operatorname{Var}[x_i(t+\Delta) - x_i(t)] = \frac{2m_i}{\Delta}\sum_{j\geq 1}\frac{w_{ij}}{\omega_j^2}[1 - \cos(\omega_j\Delta)].$$

The observable has action units and is independent of preparation time t. Set $B_i = \sum_j w_{ij}/\omega_j^2$, with units of length squared. Then

$$0 \leq \mathsf{h}_i(\Delta) \leq \frac{4m_i B_i}{\Delta} \longrightarrow 0 \quad (\Delta \to \infty).$$

At short windows, $\mathsf{h}_i(\Delta) = m_i C_i(0)\Delta + O(\Delta^3)$. The finite cosine correlation has persistent memory; its ordinary improper Green–Kubo integral need not exist. The finite-window formula above supplies the relevant limit directly, without replacing the Hamiltonian dynamics by a reversible dissipative generator.

3. Centre motion and the two-body receiver (C047)

An independent random constant centre velocity with variance σ_0^2 adds $m_i\sigma_0^2\Delta$ to h_i. This follows from $x_i(t+\Delta)-x_i(t) = V_0\Delta+$ internal displacement. The velocity ensemble is stationary; for nonzero centre motion the absolute-position ensemble need not be. A deterministic centre velocity contributes zero to the variance and can be removed by a change of frame.

For two particles of masses m, M_r coupled by a spring $k > 0$, put $\mu = mM_r/(m + M_r)$ and $\omega^2 = k/\mu$. At zero total momentum, $x_1 = R + [M_r/(m + M_r)]r$, and assign relative energy E with uniform phase. The tagged response is

$$\mathsf{h}_1(\Delta) = \frac{2mE}{k\Delta}\left(\frac{M_r}{m + M_r}\right)^2 [1 - \cos(\omega\Delta)].$$

The spring is a genuine momentum receiver: $m\ddot{x}_1 = -kr$ and $M_r\ddot{x}_2 = kr$. Its reversible exchange produces the oscillatory response rather than the exponential memory law supplied by A02's refreshed bath. Scaling every E_j by a^2, $0 < a \leq 1$, scales the entire response by a^2 while preserving the equations, conservation laws and speed margin.

4. What the large-receiver test must change (C047)

For a sequence of receivers and windows $\Delta_N \to \infty$, the bound $\mathsf{h}_{i,N}(\Delta_N) \leq 4m_{i,N}B_{i,N}/\Delta_N$ shows that a positive limiting response requires this upper bound to stay positive. In particular, uniform control of $m_{i,N}B_{i,N}$ forces a zero limit. Soft modes can enlarge $B_{i,N}$, but their weights depend on the measured coordinate and the preparation. This is the next selection test.

The finite model yields no physical-time attraction from the invariant preparation: the observable already has no t dependence. Next specify an increasing spring network, its energy allocation and tagged spectral measure, then compare taking receiver size and observation duration to infinity in opposite orders. Derive any surviving constant from these mechanical inputs. Mesh refinement of the same smooth trajectories is a third, distinct limit.

5. An increasing periodic chain (C048)

Take odd $N \geq 3$ equal masses m on a ring, with spring energy $\frac{k}{2}\sum_{i=0}^{N-1}(x_{i+1} - x_i)^2$ (indices modulo N). Fix the centre position and momentum. Give every real internal normal mode energy $\Theta > 0$ and independent uniform phase. Here Θ is a preparation energy, and total energy is $(N - 1)\Theta$. The mode frequencies and tagged covariance are

$$\omega_j = \Omega\sin(\pi j/N), \quad \Omega = 2\sqrt{k/m}, \qquad C_N(t) = \frac{\Theta}{mN}\sum_{j=1}^{N-1}\cos(\omega_j t).$$

The real sine/cosine eigenvectors have equal mode energies, so their paired squared components sum to $2/N$ at each site. Applying §2 gives

$$h_N(\Delta) = \frac{2\Theta}{N\Delta}\sum_{j=1}^{N-1}\frac{1 - \cos(\omega_j\Delta)}{\omega_j^2}.$$

For each fixed Δ, this Riemann sum converges to

$$h_\infty(\Delta) = \frac{2m}{\Delta}\int_0^\Omega \frac{1 - \cos(\omega\Delta)}{\omega^2}\rho(\omega)\, d\omega, \quad \rho(\omega) = \frac{2\Theta}{\pi m\sqrt{\Omega^2 - \omega^2}}.$$

This is a statement about limits of tagged covariances and increment variances; it does not require a stationary absolute-position law for the infinite chain. The density is integrable at its upper edge and continuous at zero. Splitting at any fixed $0 < a < \Omega$, the contribution from $[a,\Omega]$ is $O(1/\Delta)$. In the lower interval substitute $y=\omega\Delta$. Dominated convergence, using $(1-\cos y)/y^2 \in L^1(0,\infty)$ and bounded ρ on $[0,a]$, gives

$$\lim_{\Delta\to\infty} h_\infty(\Delta) = \pi m\rho(0) = \frac{2\Theta}{\Omega} = \Theta\sqrt{m/k}.$$

The integral $\int_0^\infty (1-\cos y)y^{-2}dy = \pi/2$ follows by integration by parts and the Dirichlet sine integral. Thus

$$\lim_{\Delta\to\infty}\lim_{N\to\infty} h_N(\Delta) = 2\Theta/\Omega, \qquad \lim_{N\to\infty}\lim_{\Delta\to\infty} h_N(\Delta) = 0.$$

A joint regime is also explicit. Write $f(z) = 2(1-\cos z)/z^2 = 2\int_0^1 (1-r)\cos(zr)\,dr$ with $f(0)=1$. Then $|f'| \le 1/3$. The integrand $\Theta\Delta f(\Omega\Delta\sin\pi x)$ is Lipschitz with constant at most $\pi\Theta\Omega\Delta^2/3$. A left Riemann sum error is at most that constant divided by $2N$; removing its zero mode contributes $\Theta\Delta/N$. Therefore

$$|h_N(\Delta) - h_\infty(\Delta)| \le \frac{\pi\Theta\Omega\Delta^2}{6N} + \frac{\Theta\Delta}{N}.$$

At fixed m,k,Θ, $\Omega\Delta_N \to \infty$ and $(\Omega\Delta_N)^2/N \to 0$ give the positive joint limit $2\Theta/\Omega$. The action scale is selected by the input energy and spring timescale.

6. Energy and a common speed ceiling (C049)

The positive-plateau preparation has extensive total energy. Its phase support also contains increasingly large tagged velocities. At site zero use the standard real Fourier basis. Only the $(N-1)/2$ cosine modes contribute; each has maximal velocity contribution $2\sqrt{\Theta/(mN)}$. Phases can align, so the exact support maximum is

$$\sup_\phi |v_0| = (N-1)\sqrt{\Theta/(mN)}.$$

Every neighbourhood of the maximizing phases has positive preparation measure. A common ceiling $|v_i| \le c$ for all supported trajectories thus requires $\Theta_N \le mc^2N/(N-1)^2 = O(1/N)$. This is a necessary condition from one site; no sufficiency for all sites is inferred from that maximum.

Both bounded total energy and this necessary speed condition close the response uniformly in the window. Indeed, $1-\cos z \le z^2/2$ gives $h_N(\Delta) \le \Theta_N\Delta$. The identity $\sum_{j=1}^{N-1}\csc^2(\pi j/N) = (N^2-1)/3$ gives

$$h_N(\Delta) \le \frac{4\Theta_N(N^2-1)}{3N\Omega^2\Delta}, \qquad \sup_{\Delta>0} h_N(\Delta) \le \frac{2\Theta_N}{\Omega}\sqrt{\frac{N^2-1}{3N}}.$$

The trigonometric identity follows by differentiating $\sum_{j=0}^{N-1}\cot(x+\pi j/N) = N\cot(Nx)$ and subtracting the $j=0$ singularity as $x\to 0$. The last inequality uses $\min(A\Delta, B/\Delta) \le \sqrt{AB}$. For $\Theta_N = O(1/N)$ the uniform bound tends to zero. Thus this explicit positive reservoir plateau uses a different preparation class from a family with a common strict particle-speed ceiling. Its dependence on Θ also leaves preparation-independent action selection open.

7. Source input and reproduction

Ford–Kac–Mazur, printed p. 505, (2)–(5), supplies the matrix harmonic propagator. Their canonical preparation excludes zero eigenvalues at that stage; ours fixes the centre and uses compact fixed-energy phase tori. Zwanzig, p. 219, (21)–(24), supplies a complementary reduced-description route: eliminate harmonic bath coordinates to obtain a cosine memory kernel. That friction kernel differs from the tagged velocity covariance calculated here. Its continuum replacement must be propagated through the tracer equation before identifying a displacement coefficient.

B23 records the established ingredients, derived finite-network consequence and bounded reading coverage. Historical modal, two-body and three-body checks accompany C047's proof. C047 has coordinator proof review and a sequential Luna-low prior-art audit. Its limits concern the stated observable and preparation class.

B24 audits §§5–6 against Ford–Kac–Mazur's cyclic Fourier construction and its finite-sum/infinite-chain distinction, printed p. 506, (12)–(24). The tagged density, plateau and speed-support bounds are derived here with fixed-energy phases. Six exact identities, four finite ring spectrum/cosecant comparisons and four Riemann-bound tests were recorded before the hard no-Python-verification rule; the handoff records that history. The new uncommitted verification script and output were withdrawn under the rule. Current mathematical verification uses the written proofs and source review. The next mechanism must retain bounded velocities while supplying low-frequency response; its preparation dependence remains a separate selection test.

8. What information survives a cut?

The fixed-energy two-body receiver gives an exact cut-state countertest: position-only stationary conditional kernels fail composition, while retaining momentum restores it. Independently refreshing direction at every inserted cut preserves one-time position marginals but freezes terminal position as the mesh shrinks. These are C062–C063; the full maps and proof are in the cut-state note, with the B33 audit.

With $q = x - X$, $\mu = mM/(m + M)$, $\omega = \sqrt{k/\mu}$ and $A = \sqrt{2E/k}$, uniform phase on the energy ellipse gives the conditional one-segment positions

$$q' = q \cos \omega t \ \pm \sqrt{A^2 - q^2} \sin \omega t$$

with equal weights. Thus the full mean after $s + t$ is $q \cos \omega(s + t)$, whereas independent conditional composition gives $q \cos \omega s \cos \omega t$. The missing interface variable is $p_s = \mu \dot{q}_s$ in $q_{s+t} = q_s \cos \omega t + p_s \sin \omega t/(\mu\omega)$. It has momentum units. Keeping this full internal state composes exactly; marginalizing the true joint path law also preserves all cut restrictions.

For N independent resets at intervals T/N, direct iteration gives

$$\mathbb{E}Q_N = q[\cos(\omega T/N)]^N,$$
$$\mathbb{E}Q_N^2 = \frac{A^2}{2} + \left(q^2 - \frac{A^2}{2}\right)[\cos(2\omega T/N)]^N.$$

The first limit is q and the second is q^2, proving terminal conditional L^2 convergence to the starting position. Under the original stationary preparation the reset two-time correlation tends to $A^2/2$, while the true value is $(A^2/2) \cos \omega T$. A cut that only observes the motion performs no reset and leaves the true correlation unchanged.

The normalized orbital action is $J = (2\pi)^{-1} \oint p\, dq = E/\omega$: integrating $\mu\omega A^2 \sin^2 \phi$ over a cycle proves the formula. Uniform-phase covariance also has square-root determinant E/ω. Cooling $E \downarrow 0$ at fixed Hamiltonian closes both, with exact phase-state composition intact. R04 next retains tagged momentum while eliminating an independent receiver mode, testing the additional memory exposed by Zwanzig's reduced equation.